

Income Inequality and Poverty

ECON 204 – Contemporary Economic Issues

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Learning Objectives

- How to define and measure **income inequality** and **poverty**.
- How to analyze the evolution of inequality and poverty by individual characteristics in **Canada**.
- How to compare Canada with other countries.
- How to analyze the comovement between inequality, poverty, and **real per capita GDP**.

Key terms (1/3)

- **Decile:** When the population is divided into ten equal groups ordered with respect to a variable, the groups are called deciles. For example, the first income decile is one tenth (10%) of the population with the lowest income.
- **Economic growth:** This is the growth rate of the real per capita GDP.
- **Economic mobility:** This characterizes the ability for individuals to move from one part of the income distribution to another.
- **Gini coefficient:** This is a global measure of inequality. It goes from 0 (perfect equality) to 1 (perfect inequality).

- **Income share:** This is the proportion of total income that goes to an individual or a group. For example, if the income share of females is 55%, it means that 55% of the total income is earned by women.
- **Lorenz curve:** This is a line chart that illustrates the cumulative income share of the population as a function of the proportion of the population.
- **Low income measure (LIM):** This is equal to half the median of income. It is one of the measures of the poverty line. Individuals with income less than the LIM are considered below the poverty line.

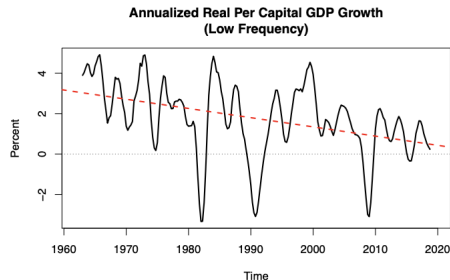
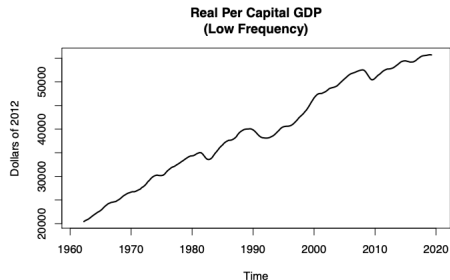
- **Market basket measure (MBM):** This is the value of a basket of goods used in one definition of the poverty line. Individuals with income less than the MBM are considered below the poverty line.
- **Median:** This is a number that separates the distribution of a variable in two equal groups.
- **Per capita:** This is a Latin word that means “by head”.
- **Quintile:** When the population is divided into five equal groups ordered with respect to a variable, the groups are called quintiles.

- Economic growth is defined as the growth rate of the **real per capita GDP** defined as

$$\text{Real per capita GDP} = \frac{\text{Real GDP}}{\text{Population}}.$$

- Since GDP is also a measure of total income, we also call it **average real per capita income**.
- To obtain real per capita GDP, download:
 - the real GDP series and
 - the population series.
- Be careful with units: population is in *units* and GDP is in *millions*; express population in millions before dividing.

Real per capita GDP and its growth (illustrations)



- Low-frequency component of real per capita GDP (left) and annualized low-frequency real per capita GDP growth (right).
- Real per capita GDP rises from about \$20,000 to about \$55,000 (2012 dollars) between 1961 and 2020; growth rate declines on average.

Why growth is not enough

- The rise in real per capita GDP implies Canadians get richer on average and standards of living improve on average.
- But it does **not** imply that the standard of living of *every* Canadian has improved.
- This lecture focuses on the **allocation** of real GDP/income across individuals, how it evolves, and how it relates to growth.

Quintiles: idea and construction

- Quintiles divide the population into five equal groups (20% each), ordered by the variable of interest.
- The lowest quintile contains the 20% with the smallest values; the fifth quintile contains the 20% with the highest values.
- Which individuals fall into each quintile depends on the variable used (e.g., age vs expenditure).

Expenditure and age by individual

Individual	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Age	19	23	45	18	64	55	34	23	22	44	70	61	49	21	43
Expenditure	100	210	80	30	312	190	201	236	341	34	40	101	170	190	343

Age quintiles (sorted by age)

	First	Second	Third	Fourth	Fifth
Individuals	4, 1, 14	9, 2, 8	7, 15, 10	3, 13, 6	12, 5, 11
Age	18, 19, 21	22, 23, 23	34, 43, 44	45, 49, 55	61, 64, 70
Expenditure	30, 100, 190	341, 210, 236	201, 343, 34	80, 170, 190	101, 312, 40

Measuring inequality: income by quintile

- A simple way to summarize inequality is to compare the **average income** across quintiles and the **share of total income** held by each quintile.
- If higher-income quintiles capture a much larger income share, the distribution is more unequal.

Average income by quintile in 2018

Quintile	Average Household Income (dollars)	Share of Total Household Income (%)
First Income Quintile	23,537	5.90
Second Income Quintile	49,137	12.40
Third Income Quintile	69,943	17.60
Fourth Income Quintile	91,798	23.10
Fifth Income Quintile	162,231	40.90

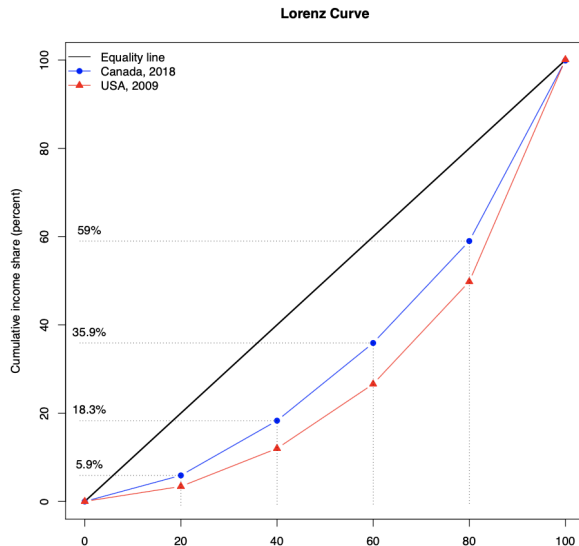
Cumulative income share in Canada in 2018

- The cumulative share of income is the share of income of the first quintile (bottom 20%), the first two quintiles (bottom 40%), the first three (bottom 60%), and so on.

Group	Cumulative Income Share (%)
The bottom 20%	5.90
The bottom 40%	18.30
The bottom 60%	35.90
The bottom 80%	59.00
The bottom 100%	99.90

- The Lorenz curve plots cumulative income share (y-axis) as a function of cumulative households (x-axis).
- Perfect equality implies: bottom 20% gets 20%, bottom 40% gets 40%, etc. (the 45-degree equality line).
- The farther the Lorenz curve is from the equality line, the more unequal the distribution.

Lorenz curve: Canada (2018) and USA (2009)



Lorenz curve: Canada (2018) and USA (2009)

- Lorenz curve with equality line.
- Coordinates shown for Canada: 5.9%, 18.3%, 35.9%, and 59%.
- Conclusion: USA (2009) allocation is more unequal than Canada (2018).

The Gini coefficient

- The Gini coefficient is based on the Lorenz curve.
- It is the area between the equality line and the Lorenz curve divided by the area under the equality line.
- The area under the equality line is 0.5, so the Gini coefficient is **twice** the area between the two curves.
- It equals 0 when the Lorenz curve is the equality line, and increases as inequality rises.

Gini coefficient with quintiles

- Let $h_i = (i/5 - s_i)$ for $i = 1, 2, 3, 4$, be the distance between the equality line and the Lorenz curve at 20%, 40%, 60% and 80%, where s_i is the cumulative income share of quintiles 1 to i .
- Then the Gini coefficient is:

$$\text{Gini} = 0.40 \times (h_1 + h_2 + h_3 + h_4) = 0.8 - 0.4 \times (s_1 + s_2 + s_3 + s_4).$$

- With quintiles, the maximum Gini is 0.8 (to reach 1, we need more than 5 groups).

Background: geometry intuition

- Between two adjacent points on the Lorenz curve, the area between the Lorenz curve and the equality line is a trapezoid.
- If the trapezoid has bases a and b and height c , its area is $(a + b) c / 2$.
- In our setup: a and b are successive distances (h_i 's), and $c = 0.20$ (20% of the population).
- Add trapezoid areas across segments and multiply by 2 to get the Gini coefficient.

Gini coefficient in general (n groups)

- If we split the population into n equal groups, define $h_i = (i/n - s_i)$ for $i = 1, \dots, n-1$, where s_i is the cumulative income share of groups 1 to i .

- Then:

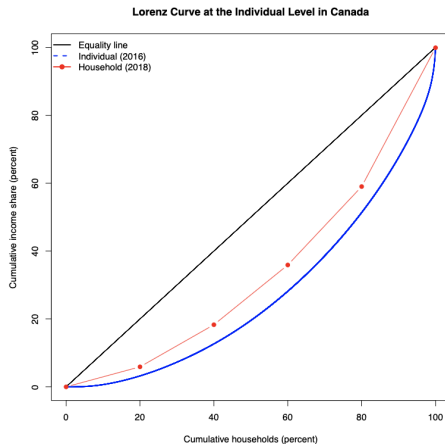
$$\text{Gini} = \frac{2}{n} \sum_{i=1}^{n-1} h_i.$$

- Replacing h_i by $i/n - s_i$ yields:

$$\text{Gini} = \frac{n-1}{n} - \frac{2}{n} \sum_{i=1}^{n-1} s_i.$$

- The maximum is $(n-1)/n$, which approaches 1 as n becomes large.

Lorenz curve: individual vs household

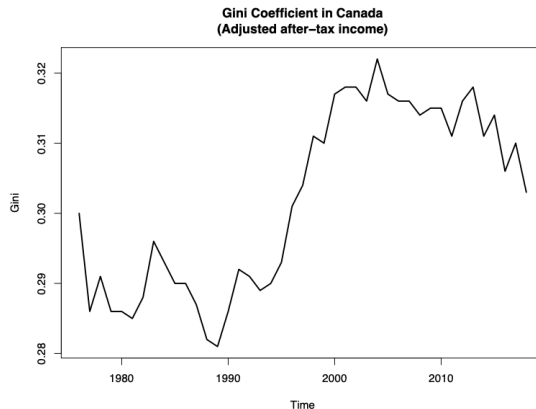


- Lorenz curve at individual level (2016) and household level (2018), with equality line.
- More groups $\Rightarrow$ smoother curve; inequality is higher at the individual level.

The evolution of the Gini coefficient in Canada

- Statistics Canada publishes an annual series of the Gini coefficient starting in 1976.
- The Gini coefficient is based on **household income adjusted for family size**, using adjusted after-tax income (income from all sources including government transfers less income tax).
- It was roughly constant from 1976 to the end of the 1980s, increased from the early 1990s to the early 2000s, then decreased onward.
- The business cycle affects inequality (e.g., increases during recessions of 1982 and 1990).

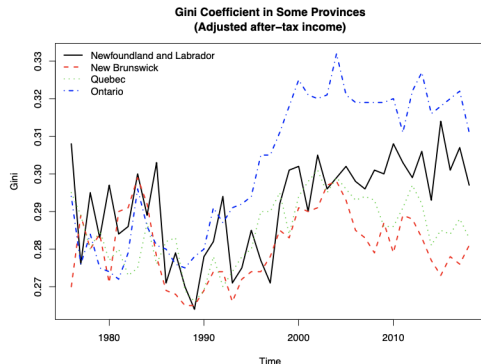
Gini coefficient in Canada



- Time series: Gini coefficient in Canada (adjusted after-tax income), 1976–2018.
- Stable through late 1980s; rises in early 1990s to early 2000s; then declines.

Gini coefficient in some provinces

- Ontario is the province where inequality increased the most over the period; by 2018, New Brunswick and Quebec had the least inequality among the four shown.

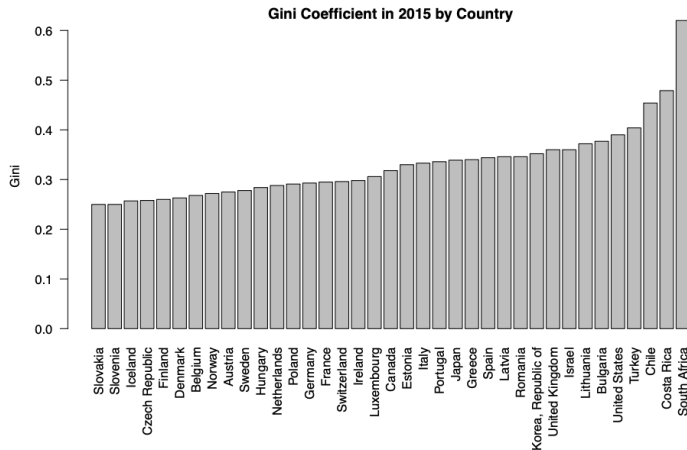


- Gini coefficient (adjusted after-tax income) for Newfoundland and Labrador, New Brunswick, Quebec, Ontario.

Comparing the Gini coefficient across countries

- To compare across countries, coefficients should come from the **same source** and be computed consistently.
- A bar chart for 2015 shows Canada in the middle among the shown countries; the United States is among the most unequal; Slovakia is among the most equal; South Africa is the most unequal in the chart.

Comparing the Gini coefficient across countries



- Bar chart: Gini coefficient in 2015 by country (selected OECD/non-OECD).

Exercises: measuring the Gini coefficient

- 1. Consider the following table of average income by quintile. Compute the Gini coefficient.

Quintile	Average Income (dollars)
First Income Quintile	342
Second Income Quintile	642
Third Income Quintile	844
Fourth Income Quintile	990
Fifth Income Quintile	1,053

Exercises: measuring the Gini coefficient (continued)

- 2. Consider the following table of average income by decile. Compute the Gini coefficient.

Decile	Average Income (dollars)
First Income Decile	114
Second Income Decile	288
Third Income Decile	605
Fourth Income Decile	621
Fifth Income Decile	1,012
Sixth Income Decile	1,023
Seventh Income Decile	1,101
Eighth Income Decile	1,434
Nineth Income Decile	1,925
Tenth Income Decile	1,934

Exercise solution (deciles): shares and cumulative shares

- Income share of decile i is average income in decile i divided by the sum of average incomes.
- Example: for the first decile,

$$S_1 = \frac{114}{114 + 288 + 605 + 621 + 1,012 + 1,023 + 1,101 + 1,434 + 1,925 + 1,934} = 1.13\%.$$

- Compute cumulative shares, then apply the general formula with $n = 10$:

$$\text{Gini} = \frac{n-1}{n} - \frac{2}{n}(s_1 + s_2 + \cdots + s_9).$$

Decile shares table (rounded at the end)

Decile	Avg. Income	Income (%)	share	Cumulative shares (%)
First Income Decile	114	1.13		1.13
Second Income Decile	288	2.86		4.00
Third Income Decile	605	6.02		10.01
Fourth Income Decile	621	6.17		16.19
Fifth Income Decile	1,012	10.06		26.25
Sixth Income Decile	1,023	10.17		36.42
Seventh Income Decile	1,101	10.95		47.37
Eighth Income Decile	1,434	14.26		61.63
Ninth Income Decile	1,925	19.14		80.77
Tenth Income Decile	1,934	19.23		100.00

- Income inequality is particularly problematic when it implies that a large proportion of the population is in a state of poverty.
- An household is considered poor if its income falls below the **poverty line**.
- Two measures used by Statistics Canada:
 - the **market basket measure (MBM)**, and
 - the **low income measure (LIM)**.

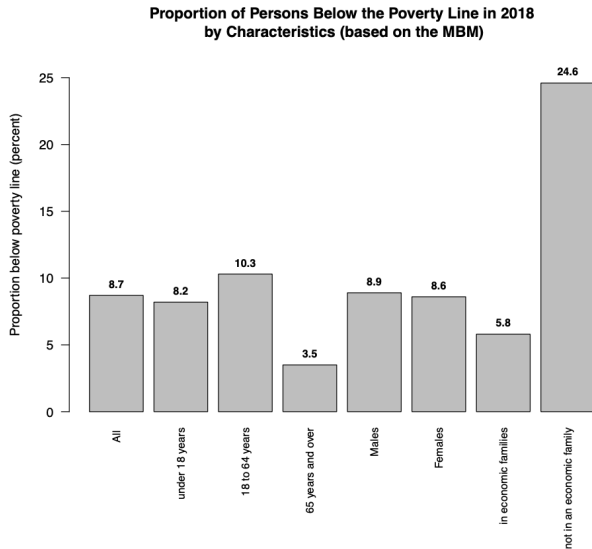
The market basket measure (MBM)

Definition

The Market Basket Measure (MBM) is based on the cost of a specific basket of goods and services representing a modest, basic standard of living. It includes the costs of food, clothing, shelter, transportation and other items for a reference family. These costs are compared to the disposable income of families to determine whether or not they fall below the poverty line.

- A household is below the poverty line if it cannot afford the “modest and basic standard of living” determined by the basket.
- The table includes the proportion of persons below the poverty line by characteristics starting in 2006.

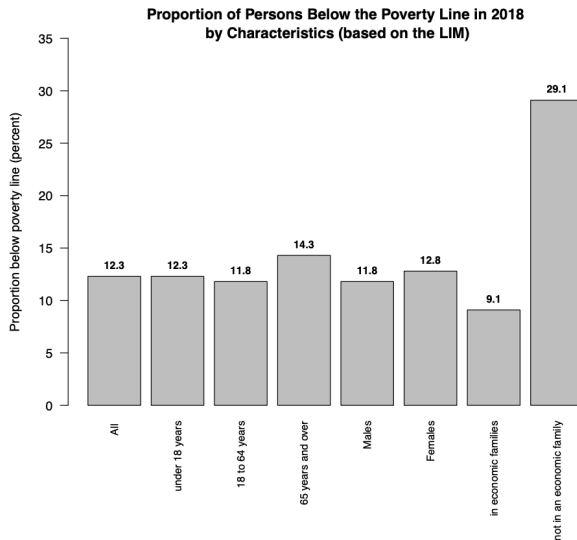
Proportion below poverty line in 2018 (MBM)



The low income measure (LIM)

- Another definition of poverty line is the low income measure (LIM).
- It is equal to half the adjusted median household income (adjusted for household size).
- Median household income in 2018 was \$61,400, so households with income less than \$30,700 are below the poverty line (before adjusting for household size).
- The LIM series is based on adjusted after-tax income.

Proportion below poverty line in 2018 (LIM)

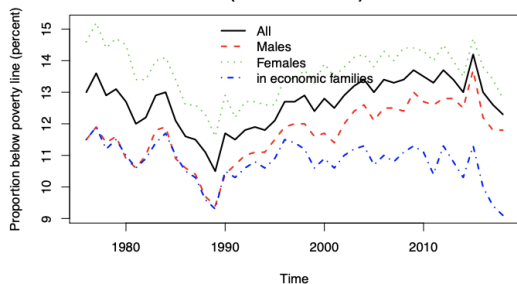


Poverty over time in Canada (LIM)

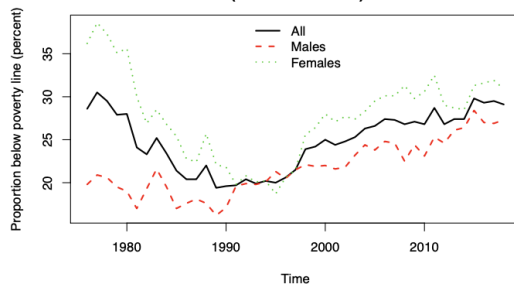
- Evolution of poverty from 1976 to 2018 based on LIM.
- Left graph: all persons, males, females, persons in economic families.
- Right graph: all persons not in family, males not in family, females not in family.
- Overall: poverty decreased from 1976 to 1990 and increased after 1990; persons in family were quite stable.
- Females were systematically more likely to be below the poverty line than males, but the gap narrowed by 2018.

Poverty over time

**Proportion of Persons Below the Poverty Line by Characteristics
(based on the LIM)**



**Proportion of Persons Not in Family Below the Poverty Line
(based on the LIM)**



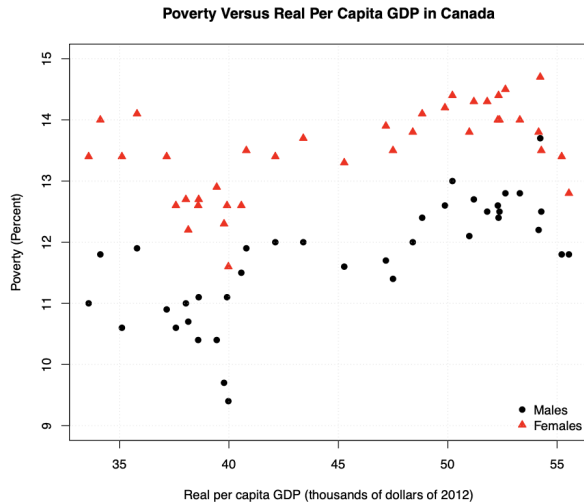
Important: poverty vs inequality

- Is poverty related to inequality? **Not necessarily.**
- Everyone could be poor $\Rightarrow$ poverty high but inequality low.
- Everyone could be above the poverty line $\Rightarrow$ poverty low but inequality could still be high.
- Poverty focuses on the bottom of the distribution; inequality focuses on dispersion across the whole distribution.

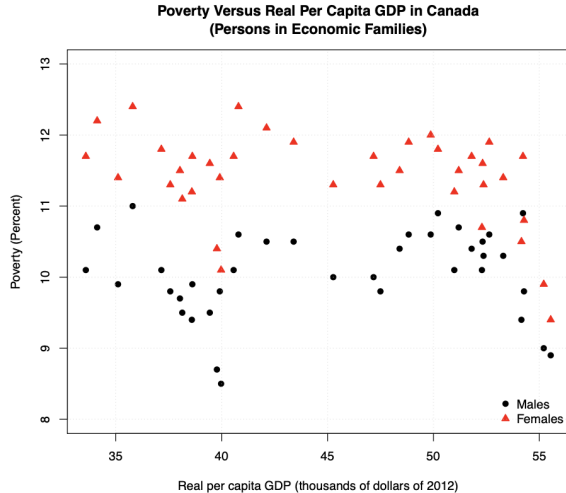
Exercise: poverty versus per capita GDP in Canada

- You need:
 - proportion of persons below the poverty line (LIM) by characteristics,
 - annual real GDP series, and
 - annual population series.
- Analyze the relationship between poverty and real per capita GDP by characteristics.
- Use scatter plots and compare:
 - males vs females,
 - males vs females in economic families,
 - males vs females not in economic families.

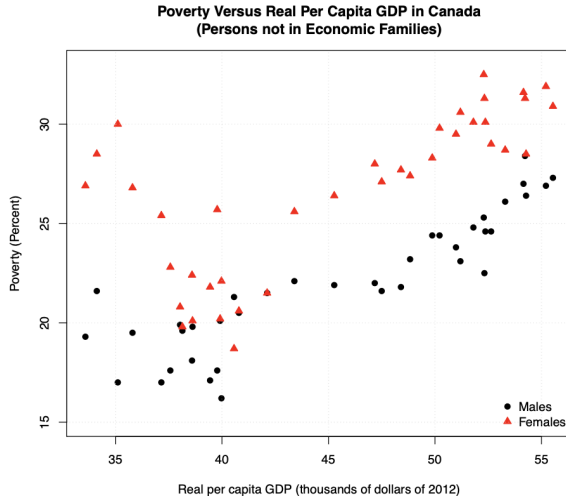
Exercise solution: scatter plots



Poverty vs GDP: persons in economic families



Poverty vs GDP: persons not in economic families



Sources of inequality

- **Inequality in skills:** Highly demanded skills command higher pay (e.g., skilled NHL hockey players; coding skills).
- **Inequality in how skills are valued:** Income differences depend on how society values goods/services and on demand vs supply of skills.
- **Inequality in education:** Higher education leads to higher income; education gaps contribute to income inequality.
- **Technological advances:** Often increase the value of certain skills/education (e.g., IT boom), affecting inequality.
- **Value of education:** Anything that raises the return to education amplifies how education inequality translates into income inequality.

Median annual earnings by education (2016)

	High school diploma	Apprenticeship certificate	College diploma	Bachelor's degree
Women	\$43,254	\$38,230	\$48,599	\$68,342
Men	\$55,774	\$72,955	\$67,965	\$82,082

- More educated workers earn more: about \$25,000 gap between high school diploma and bachelor's degree.
- Policies that reduce education inequality can also reduce income inequality.

The effect of inequality on growth

- Explaining the effect of inequality on growth requires theory and careful empirical work.
- Different economists reach different conclusions using available data.
- Inequality may affect growth through channels such as:
 - **investment** (capital accumulation),
 - **education** (worker productivity),
 - **taxation/redistribution** (incentives),
 - **political stability** (investment incentives),
 - **criminality** (resources wasted on prevention rather than production).
- Understand the intuition behind channels and the difficulty of measuring their relative importance.

- Economic mobility is the ability for individuals to move from one part of the income distribution to another (e.g., from the first quintile to the second or third).
- A transition matrix can summarize mobility.
- Example patterns:
 - If parents are in the bottom quintile, children have 42% chance of being in the bottom decile, 23% chance of being in the second, and 6% chance of being in the top decile.
 - If parents are in the third quintile, children have almost equal chance to be in any quintile.
- A key advantage of a highly mobile economy is that everyone has a more equal chance of being successful.

- Growth raises average living standards, but distribution matters for who benefits.
- Inequality can be summarized using income shares, Lorenz curves, and the Gini coefficient.
- Poverty depends on the poverty line definition (MBM vs LIM) and varies by characteristics and over time.
- Inequality can interact with growth and mobility through multiple channels.